Methodology

Forecasting the Realized Volatility of the Rare Earth ETF (REMX) with News and Social Media Sentiment from GDELT

Master's Thesis · BSE Financial Economics
Aleix Cuevas · Niklas Biemel · Nicolas Koutalakis

Overview and rationale

This document sets out the empirical methodology for our master's thesis, which combines two complementary strands of the literature on news-driven volatility forecasting. From **Guidolin and Pedio (2020)** we borrow the family of GARCH-X specifications augmented with tone and article-growth regressors, originally developed for the FTSE 100 during the Brexit period. From **Dutta et al. (2025)** we borrow the choice of the VanEck Rare Earth/Strategic Metals ETF (REMX) as the target asset and the range-based realized variance estimators (Parkinson, Rogers-Satchell) to construct a daily volatility proxy from OHLC data. Our empirical contribution is the combination of these two frameworks on a previously unexplored asset: rare earth equities, using a publicly available and replicable sentiment source (GDELT) in place of Dutta et al.'s proprietary Thomson Reuters MarketPsych Indices (TRMI).

The pivot from physical rare-earth metal prices (Nd, Pr, Tb, Dy) to the REMX ETF is motivated by the stickiness and OTC nature of physical REE prices, which violate the IID innovation assumptions required by GARCH-type models. The REMX, traded on NYSE Arca since October 2010 with full OHLC data available, provides a clean target for conditional volatility modelling while remaining the standard proxy used by the recent REE finance literature (Bouri et al. 2021; Song et al. 2021; Hanif et al. 2023; Madaleno et al. 2023; Dutta et al. 2025).

Our sample period runs from **1 April 2015 to 31 March 2026**, providing approximately 2,750 trading days. This window starts after the operational stabilisation of GDELT 2.0 (launched February 2015) and covers a sequence of major events relevant to REE markets, including the US-China trade war, the COVID-19 pandemic, the Russia-Ukraine conflict, and China's April 2025 rare earth export controls.

The methodology is organised into nine sequential blocks. Each block is self-contained and can be executed and validated independently before moving to the next. Blocks 1-3 prepare the data; Block 4 partitions the sample; Blocks 5-7 estimate and evaluate the models; Blocks 8-9 provide robustness checks and optional extensions.

Block 1 — Construction of the volatility target

This block builds the dependent variable: a daily realized variance series for the REMX ETF, using range-based estimators that exploit the OHLC structure of daily price data.

Step 1.1 — Obtain OHLC data for the REMX

Download daily Open, High, Low, and Close prices for the VanEck Rare Earth/Strategic Metals ETF (ticker: REMX) from 1 April 2015 to 31 March 2026. Sources: Bloomberg, Yahoo Finance, or Refinitiv. Verify that the series is adjusted for splits and dividends and that there are no missing trading days or anomalous outliers caused by uncorrected corporate actions.

Step 1.2 — Compute daily log returns

Define close-to-close log returns:

$$r_t = ln(C_t / C_{t-1})$$

This series is the input for the conditional-mean equation of the GARCH models estimated in Block 5.

Step 1.3 — Compute Parkinson realized variance

For each trading day, calculate the Parkinson (1980) range-based estimator:

$$RV_t^{Park} = (1 / 4 \ln 2) \times (\ln H_t - \ln L_t)^2$$

The factor 1/(4 ln 2) ensures unbiasedness under the assumption of a driftless geometric Brownian motion. This is the primary RV measure used by Dutta et al. (2025) and serves as our main proxy for the latent conditional variance.

Step 1.4 — Compute Rogers-Satchell realized variance

For each trading day, calculate the Rogers and Satchell (1991) estimator:

$$RV_t^{RS} = \ln(H_t/O_t) \cdot \ln(H_t/C_t) + \ln(L_t/O_t) \cdot \ln(L_t/C_t)$$

Unlike the Parkinson estimator, Rogers-Satchell is robust to non-zero drift in the price process, making it more appropriate when the underlying asset exhibits a persistent trend within the day. It serves as our secondary RV measure for robustness checks.

Step 1.5 — Diagnostic checks on the RV series

Compute descriptive statistics (mean, standard deviation, skewness, kurtosis) and plot both RV series over time. Verify that volatility spikes coincide with known REE market events: the April 2025 China export controls should appear as a clear peak; the March 2020 COVID-19 shock should be visible across asset classes. If the spikes do not match documented events, return to the data sourcing step.

Block 2 — Construction of the sentiment regressors

This block builds the explanatory variables: daily series of tone and article growth derived from GDELT, aligned to the REMX trading calendar.

Step 2.1 — Apply the final filter to the GDELT cache

Starting from the cached article-level dataset built in the BigQuery extraction stage, apply the chosen topic filter to isolate REE-relevant articles. Based on the quality validation performed earlier, the concept co-occurrence filter (Filter 2) with a threshold of three distinct concepts provides the best balance between precision and recall.

Step 2.2 — Aggregate to daily frequency

For each calendar day t, compute the mean tone across all relevant articles and the total article count. Optionally, also aggregate the positive score, negative score, and polarity components of V2Tone as separate daily means for later sensitivity analyses.

Step 2.3 — Align the sentiment calendar with the REMX calendar

The REMX trades only on US business days (approximately 252 per year), whereas GDELT records articles every calendar day. Following Guidolin and Pedio (2020), accumulate the sentiment from non-trading days (weekends and holidays) into the next available trading day, so that the Monday-morning observation incorporates the news flow of the preceding weekend. This treatment respects the assumption that weekend information is priced at the next market opening.

Step 2.4 — Construct the article growth variable

Following Guidolin and Pedio (2020), define:

$$Art\_growth_t = (n\_articles_t - n\_articles_{t-1}) \,/\, n\_articles_{t-1}$$

This captures the relative change in media attention between consecutive trading days. Handle the edge case $n\_articles_{t-1} = 0$ by setting Art_growth to missing.

Step 2.5 — Rescale the sentiment variables

Multiply $tone_t$ by 100 to align its scale with the conditional variance of percentage returns, as done by Guidolin and Pedio. This rescaling improves the numerical stability of the maximum likelihood optimisation. Consider similar rescaling for Art_growth if convergence problems arise.

Step 2.6 — Outlier treatment

Identify trading days with extreme tone or article growth values (typically beyond ±3 standard deviations). Choose between removal, winsorisation (truncation at the 1st and 99th percentiles), or retention. Document the choice explicitly and report results under at least two treatments as a robustness check.

Step 2.7 — Diagnostic checks on the sentiment series

Plot tone and article growth over time. Verify that peaks in article growth coincide with known REE news events. Compute the pairwise correlation between sentiment variables and the RV series as a preliminary indication of explanatory power.

Block 3 — Joint exploratory analysis

Step 3.1 — Descriptive statistics for all series

Produce a single table reporting mean, standard deviation, skewness, kurtosis, minimum, and maximum for: the daily return r_t, both RV measures (Parkinson and Rogers-Satchell), the tone series, the article growth series, and their squares. This table replicates Table 1 of Dutta et al. (2025) and Table 1 of Guidolin and Pedio (2020) adapted to our setting.

Step 3.2 — Correlation matrix

Compute the pairwise Pearson correlations between r_t, r_t^2, RV_t^{Park}, RV_t^{RS}, $tone_t$, $tone_t^2$, Art_growth_t, and $Art\_growth_t^2$. Bold-face entries significant at the 5% level. This table replicates Table 2 of Guidolin and Pedio (2020).

Step 3.3 — Stationarity tests

Apply the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests to all series. Returns should be clearly stationary; RV may be highly persistent but should still reject the unit root; tone and article growth should be stationary. If any series fails to reject, apply first differencing before proceeding to the modelling stage.

Block 4 — Train/test partition

Step 4.1 — Define the in-sample and out-of-sample windows

Following Guidolin and Pedio (2020), use an approximate 70/30 split. With our sample of approximately 2,750 trading days from 1 April 2015 to 31 March 2026:

In-sample: 1 April 2015 to 31 December 2022 (approximately 1,950 trading days)
Out-of-sample: 1 January 2023 to 31 March 2026 (approximately 800 trading days)

Step 4.2 — Consider rolling-window re-estimation

For the out-of-sample exercise, the most rigorous approach is to re-estimate each model on a rolling window of the most recent N years (e.g., N = 5) and produce one-step-ahead forecasts. This is the approach used by Guidolin and Pedio. If computational cost is prohibitive, fall back to a single in-sample estimation followed by a sequence of conditional forecasts over the out-of-sample period.

Block 5 — Estimation of volatility models

This block estimates the family of GARCH-type models. All models share a common conditional mean equation $r_t = \mu + \varepsilon_t$ with $\varepsilon_t = \sigma_t z_t$, $z_t \sim N(0,1)$, and differ in the specification of the conditional variance σ_t^2.

Step 5.1 — Benchmark GARCH(1,1) without sentiment

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2$$

Estimate by maximum likelihood with Bollerslev-Wooldridge quasi-MLE standard errors. Verify covariance stationarity ($\alpha + \beta < 1$) and significance of all parameters. This is the benchmark against which all augmented models are compared.

Step 5.2 — GARCH-X with tone (Guidolin-Pedio equation 4)

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma \cdot tone_{t-1}^2$$

Tone enters squared because the theoretical assumption is that extreme news (in either direction) generates volatility; the sign of tone alone should not matter — only its magnitude.

Step 5.3 — GARCH-X with article growth

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma \cdot Art\_growth_{t-1}^2$$

Same logic applied to the article growth: jumps in media attention (in either direction) should generate volatility.

Step 5.4 — GARCHAND with asymmetric tone (Guidolin-Pedio equation 5)

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma\,(1 + d_1\, \theta) \cdot tone_{t-1}^2$$

Here $d_1 = 1$ when $tone_{t-1} < 0$, zero otherwise; θ is positive and bounded below by –1. The model captures the asymmetry by which negative news has a larger impact on subsequent volatility than positive news, analogous to the classical leverage effect. A significant and positive estimate of θ confirms this asymmetric mechanism.

Step 5.5 — GARCHAND with asymmetric article growth

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma\, d_2 \cdot Art\_growth_{t-1}^2$$

Here $d_2 = 1$ when $Art\_growth_{t-1} > 0$, and γ is restricted to be positive. The intuition is that only surges in media attention should impact volatility; declines in coverage do not carry symmetric effects.

Step 5.6 — GARCHND with regime-dependent sentiment (Guidolin-Pedio equation 7)

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma\, d_3 \cdot x_{t-1}^2$$

Here $d_3 = 1$ when $\sigma_{t-1}^2 \geq \kappa$ for some threshold κ, and x denotes either tone or article growth. The hypothesis is that sentiment matters only when conditional volatility is already elevated. Experiment with thresholds corresponding to the 75th and 90th percentiles of in-sample volatility.

Step 5.7 — EGARCH-X with tone (additional extension)

$$ln(\sigma_t^2) = \omega + \alpha(|z_{t-1}| - E|z_{t-1}|) + \gamma_L\, z_{t-1} + \beta\, ln(\sigma_{t-1}^2) + \gamma_S \cdot tone_{t-1}$$

The EGARCH (Nelson 1991) models the logarithm of the conditional variance, removing the need for positivity restrictions on parameters and allowing tone to enter linearly without the squared transformation. The leverage parameter γ_L captures asymmetry in the return innovations; γ_S captures the additional impact of news sentiment. Although not part of the original Guidolin-Pedio framework, this specification is widely used in the related literature and provides a useful comparison.

Step 5.8 — Estimate all models with robust standard errors

All models must be estimated with Bollerslev-Wooldridge (QMLE) robust standard errors. Daily financial returns exhibit fat tails that violate the conditional normality assumption underlying naive MLE inference; QMLE corrects the asymptotic variance accordingly and is the standard approach in this literature.

Block 6 — In-sample evaluation

Step 6.1 — Coefficient significance

For each augmented model, test whether the sentiment coefficient (γ in GARCH-X variants, γ_S in EGARCH-X) is significantly different from zero at the 5% level using robust standard errors. Statistical significance is a necessary but not sufficient condition for the model to be useful.

Step 6.2 — Residual diagnostics

For each model, apply the following diagnostic tests to the standardised residuals $z_t = \varepsilon_t / \sigma_t$:

- Ljung-Box test on z_t (lags 5, 10, 20): should fail to reject the null of no autocorrelation
- Ljung-Box test on z_t^2 (lags 5, 10, 20): should fail to reject
- ARCH-LM test on z_t: should fail to reject the null of no remaining ARCH effects
- Jarque-Bera normality test (informative only, not critical under QMLE)

A model that passes all these diagnostics has adequately captured the volatility structure of the data.

Step 6.3 — In-sample fit comparison

Produce a summary table with the log-likelihood, AIC, and BIC of each model. The model with the lowest AIC/BIC is the in-sample winner. Note that lower in-sample information criteria do not automatically translate into better out-of-sample forecasting; this is precisely why Block 7 is necessary.

Block 7 — Out-of-sample evaluation

This block constitutes the core empirical test of the thesis: whether the sentiment-augmented models produce conditional variance forecasts that are systematically more accurate than the GARCH(1,1) benchmark over a holdout period not used for estimation.

Step 7.1 — Generate one-step-ahead forecasts

For each day t in the out-of-sample period and each model, produce the conditional variance forecast $\sigma_{t+1|t}^2$ using information available up to time t. If the rolling-window approach is feasible, re-estimate the model parameters at each step.

Step 7.2 — Direct loss functions against the RV proxy

The Parkinson and Rogers-Satchell realized variances play the role of the observable proxy for the latent conditional variance, exactly as the intraday realized variance does in Guidolin and Pedio. Compute, for each model and each RV proxy:

$$RMSE = sqrt[\ (1/T)\ \Sigma\ (RV_{t+1} - \sigma_{t+1|t}^2)^2\ ]$$

$$MAE = (1/T)\ \Sigma\ |RV_{t+1} - \sigma_{t+1|t}^2|$$

Step 7.3 — Heteroskedasticity-adjusted RMSE (Bollerslev-Ghysels 1996)

$$HRMSE = sqrt[\ (1/T)\ \Sigma\ ((RV_t - \sigma_{t|t-1}^2) / RV_t)^2\ ]$$

The HRMSE is the standard volatility-comparison statistic in the recent literature (used by Dutta et al. 2025 as their primary measure). It normalises the forecast error by the realized variance, ensuring that periods of high and low volatility contribute proportionally to the loss.

Step 7.4 — Diebold-Mariano test against the GARCH(1,1) benchmark

For each augmented model, perform a pairwise Diebold-Mariano (1995) test against the GARCH(1,1) benchmark. A significantly negative DM statistic indicates that the augmented model has systematically smaller forecast errors than the benchmark. **This is the central statistical test of the entire thesis**: if sentiment adds genuine predictive value, it must show up here.

Step 7.5 — Out-of-sample R^2 (Campbell-Thompson 2008)

$$R^2_{OOS} = 1 - [\, \Sigma (RV_t - RV_t^{model})^2 \,] \,/\, [\, \Sigma (RV_t - RV_t^{benchmark})^2 \,]$$

A positive R^2_{OOS} indicates that the augmented model outperforms the benchmark. Accompany this statistic with the MSPE-adjusted test of Clark and West (2007), as done by Dutta et al. (2025), to formally assess whether the improvement is statistically significant.

Block 8 — Robustness checks

Step 8.1 — Repeat with the alternative RV proxy

Replicate the entire out-of-sample evaluation using Rogers-Satchell in place of Parkinson. Verify that the qualitative conclusions are unchanged: the same models should be best (or near-best) under both proxies. If the rankings differ substantially, discuss the implications and identify which proxy is more appropriate for the REMX given its observed drift characteristics.

Step 8.2 — Sensitivity to the sentiment filter

Rebuild the tone and article growth series under at least two alternative filter thresholds (e.g., n_keywords ≥ 2 and n_keywords ≥ 4) and verify that the main results carry over. This addresses concerns that the findings depend on a specific filtering choice.

Step 8.3 — Alternative train/test splits

Repeat the evaluation under 80/20 and 60/40 splits, and verify that the best-performing model is robust to the choice of split. Particularly important if the out-of-sample period happens to coincide with an atypical regime (e.g., the April 2025 China export controls).

Step 8.4 — Sub-period stability

Estimate the best-performing model separately on two equal sub-periods (e.g., 2015-2019 and 2020-2022) and verify that the sentiment coefficient is stable across them, in the sense that the confidence intervals overlap. Instability in this coefficient would indicate that the sentiment-volatility relationship has evolved over time, which is itself an interesting finding worth discussing.

Step 8.5 — Placebo sentiment test

Construct a placebo sentiment series from GDELT articles that do NOT pass the REE filter (e.g., articles tagged with completely unrelated topics such as sports or cryptocurrency). If this placebo sentiment also predicts the volatility of the REMX, the relationship discovered with the genuine REE sentiment may reflect generic news-cycle co-movement rather than REE-specific information. If it does not (the expected result), this strengthens the internal validity of the main findings considerably.

Step 8.6 — Robustness check on physical metal prices

Apply the best-performing model to the daily squared returns r_t^2 of physical REE prices (Nd, Pr, Tb, Dy from Asian Metal or SMM). Given the stickiness of these prices, expect weaker results than on the REMX, but the direction should be consistent. If sentiment predicts REMX volatility but not physical price volatility, this is an interesting finding in itself, indicating that the transmission of news to the financial market and to the physical commodity market operates on different timescales.

Block 9 — Optional extensions

Step 9.1 — Noise-injection robustness

Add Gaussian noise of increasing variance to the tone series and re-estimate the best model. The sentiment coefficient should decay gradually toward zero as the noise level rises, confirming that the model captures a genuine signal rather than a spurious artefact.

Step 9.2 — Decomposition of tone into positive and negative components

Replace the single tone variable with separate pos_score and neg_score series and re-estimate the GARCH-X specification. Tetlock (2007) finds that pessimism (negative tone) has more predictive power than optimism (positive tone). Test whether this asymmetry holds in the REE context.

Step 9.3 — Metal-specific sentiment indices

Use the GDELT keyword structure to build separate tone series for articles mentioning each of the four target metals (Nd, Pr, Tb, Dy). Test whether the volatility of the REMX is more sensitive to news about specific metals than to aggregate REE news. This is a contribution that neither Guidolin-Pedio nor Dutta et al. can make with their data structures.

Step 9.4 — Concept-level sentiment decomposition

Using the precomputed has_<concept> booleans in the cached dataset, construct tone series restricted to articles tagged with specific concepts: China-related news, supply chain news, electric vehicle news, export control news, and so on. Test which thematic dimension of news has the strongest impact on REMX volatility. The ex-ante hypothesis is that China and export-control news will dominate, particularly around April 2025, but this should be tested empirically rather than assumed.

Resulting thesis structure

The methodology described above produces the natural outline for the thesis:

1. Introduction — motivation, research question, contributions to the literature

2. Literature review — Guidolin and Pedio (2020), Dutta et al. (2025), Andersen-Bollerslev (2001), Hansen-Lunde (2005), and the broader rare-earth and sentiment literatures

3. Data — GDELT extraction and filtering, REMX OHLC, the 2015-2026 sample window

4. Methodology — Parkinson and Rogers-Satchell estimators, GDELT sentiment construction, the GARCH model suite, and the forecast evaluation framework

5. Empirical results — descriptive statistics, in-sample estimation, out-of-sample evaluation, Diebold-Mariano tests

6. Robustness — alternative proxy, alternative filters, sub-periods, placebo, physical prices

7. Discussion — interpretation of results, comparison with Dutta et al. (2025) and Guidolin-Pedio (2020), implications for the literature

8. Conclusions — implications for investors, policymakers, and future research

Key references

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2001). The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*.

Bollerslev, T. and Ghysels, E. (1996). Periodic autoregressive conditional heteroscedasticity. *Journal of Business and Economic Statistics*, 14(2), 139-151.

Campbell, J. Y. and Thompson, S. B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average? *Review of Financial Studies*, 21(4), 1509-1531.

Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291-311.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13, 253-263.

Dutta, A., Sihvonen, J., Park, D., Lucey, B., and Uddin, G. S. (2025). Impact of news and social media sentiments on rare earth investments. *Resources Policy*, 107, 105655.

Guidolin, M. and Pedio, M. (2020). Media attention vs. sentiment as drivers of conditional volatility predictions: an application to Brexit. *BAFFI CAREFIN Working Paper No. 145*, Bocconi University.

Hansen, P. R. and Lunde, A. (2005). A realized variance for the whole day based on intermittent high-frequency data. *Journal of Financial Econometrics*.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: a new approach. *Econometrica*, 59(2), 347-370.

Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61-65.

Rogers, L. C. G. and Satchell, S. E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504-512.

Tetlock, P. C. (2007). Giving content to investor sentiment: the role of media in the stock market. *Journal of Finance*, 62(3), 1139-1168.